STATISTICS CALCULATORS FOR MATH 105

Statistics Calculators for Math 105

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1: Scientific Calculator powered by DESMOS

Scientific Calculator powered by DESMOS

This is a scientific calculator powered by DESMOS, and the same calculator is also available at the bottom of each statistics calculator. Please report error to Dr. Jessica Kuang at jkuangATvcccd.edu.

2: One Variable Statistics

One Variable Statistics

This calculator computes the mean, standard deviation, and 5-number summary from a one-variable data set. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here.

Input

Type in the values from the data set separated by commas (for example, 2,4,5,8,11,2), choose between population standard deviation and sample standard deviation and click Calculate.

- Population Standard Deviation
- Sample Standard Deviation

Calculate

Output

Mean:

Minimum:

Q1:

Median:

Q3:

Maximum:

Standard Deviation:

Variance:

Interquartile Range (IQR):

Sample Size:

Scientific Calculator

3: Mean and Standard Deviation from a Frequency Table

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Mean and Standard Deviation from a Frequency Table

This calculator computes mean, standard deviation, and 5-number summary from a frequency or probability distribution table. Please report any error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here.

Input

Type in the data values and frequencies (in whole numbers or decimals) below.

data value	
frequency	

Population Standard Deviation

Sample Standard Deviation

Calculate

Output

Mean:

Standard Deviation:

Five Point Summary:

Sample Size:

Scientific Calculator

4: Binomial Probability Distribution

Binomial Probability Distribution

This calculator computes P(a ≤ x ≤ b), where x is a binomial random variable, and a and b are the lower bound and the upper bound for x respectively. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, watch a short video here.

Input

Enter the lower bound and the upper bound for the number of successes, the number of trials (n), and the probability of success (p), and then hit Calculate.

Lower Bound:	Upper Bound:	n:	p (enter a decimal):

Calculate

Output

Probability:

Scientific Calculator

5: Normal Probability Distribution

Normal Probability Distribution

This calculator computes P(a < x < b), where x is a normal random variable, and a and b are the lower bound and upper bound for x. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here.

Input

Mean:	Standard Deviation:

Output

Enter 2 out of 3 values below to either calculate the probability or a boundary value
(enter integers or decimals only)

Lower Bound:	Upper Bound:	Probability:

Calculate

Error Message:

Scientific Calculator

6: Linear Correlation and Regression

Linear Correlation and Regression

This calculator creates a scatter plot, the regression equation, r and r^2, and performs the hypothesis test for a nonzero correlation. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here.

Input

Put the independent variable's data separated by commas in the first row and the dependent variable's data separated by commas in the second row, then click Plot Points and Calculate.

Explanatory/Independent Variable (x):

Response/Dependent Variable (y):

Plot Points and Calculate | Reset

Output

Regression Equation:

r:

r^2:

- Hypothesis: $H_0: \rho = 0, H_1: \rho \neq 0$
- Hypothesis: $H_0: \rho = 0, H_1: \rho < 0$
- Hypothesis: $H_0: \rho = 0, H_1: \rho > 0$

Test Statistic (t):

p-value:

Scientific Calculator

7: Confidence Interval for a Population Proportion

Confidence Interval for a Population Proportion

This calculator creates confidence intervals for a population proportion given statistics. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here.

Input

Fill in the sample size (n), the number of successes (x), and the confidence level (CL), then hit Calculate. Write the confidence level as a decimal. For example, for a 95% confidence level, enter 0.95.

Sample Size (n):	Number of Successes (X):	Confidence Level (enter a decimal):

Calculate

Output

Point Estimate ($\hat{p}$):	Lower Bound:	Upper Bound:

Scientific Calculator

8: Sample Size to Estimate a Population Proportion

Sample Size to Estimate a Population Proportion

This calculator computes the sample size needed to estimate a population proportion. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here.

Input

Fill in the estimation for P (or choose no estimation for P), the error bound and the confidence level, then hit "Calculate " and the needed sample size will be calculated for you.

No Estimate for p Have Estimate for p	Estimate for p (enter a decimal):	Error Bound (enter a decimal):	Confidence Level (enter a decimal):

Calculate

Output

Sample Size (n):

Scientific Calculator

Estimate for p:

9: Confidence Interval for a Population Mean Given Statistics

Confidence Interval for a Population Mean Given Statistics

This calculator creates a confidence interval for a population mean given statistics. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here.

Input

Fill in the sample size (n), the sample mean ($\bar{x}$), the sample standard deviation (s), and the confidence level (CL), then click Calculate. Write the confidence level as a decimal. For example, for a 95% confidence level, enter 0.95.

σ Unknown	n:	$\bar{x}$:	CL:

Calculate

Output

	Lower Bound	Upper Bound

Scientific Calculator

10: Confidence Interval for a Population Mean Given Data

Confidence Interval for a Population Mean Given Data

This calculator creates a confidence interval for a population mean given a set of data. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here.

Input

Type in the values from the data set separated by commas, for example, 2,4,5,8,11,2. Then type in the confidence level (CL) and hit Calculate. Write the confidence level as a decimal. For example, for a 95% confidence level, enter 0.95.

Data:

Confidence Level (enter a decimal):

Calculate

Output

Sample Mean ($\bar{x}$):	Standard Deviation (s):	Lower Bound:	Upper Bound:

Scientific Calculator

11: Sample Size to Estimate a Population Mean

Sample Size to Estimate a Population Mean

This calculator calculates the sample size needed to estimate a population mean. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here.

Input

Fill in the population standard deviation (σ), the error bound (E), and the confidence level (CL). Write the confidence level as a decimal. For example, for a 95% confidence level, enter 0.95 for CL. Then hit Calculate and assuming the population is normally distributed, the necessary sample size will be shown.

Standard Deviation (σ):	Error Bound (E, enter a decimal):	Confidence Level (enter a decimal):	

Output

Calculate

Sample Size (n):	

Scientific Calculator

12: Hypothesis Test for a Population Proportion

Hypothesis Test for a Population Proportion

This calculator performs the hypothesis test for a population proportion given statistics. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here.

Input

Fill in the sample size, n, the number of successes, x, the hypothesized population proportion p_0, and indicate if the test is left tailed, <, right tailed, >, or two tailed, $\neq$. Then hit "Calculate" and the test statistic and p-Value will be calculated for you.

Sample Size (n):

Number of Success (x):

Choose the test

- $<$
- $>$
- $\neq$

Hypothesized Population Proportion (p_0):

Calculate

Output

Test Statistics (z)

p-value:

Scientific Calculator

Back to the Calculator Menu

13: Hypothesis Test for a Population Mean Given Statistics

Hypothesis Test for a Population Mean given Statistics

This calculator performs the hypothesis test for a population mean given statistics. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here.

Input

Select if the population standard deviation, σ, is known or unknown. Then fill in the sample size (n), the sample mean ($\bar{x}$), the standard deviation (s), the hypothesized population mean μ_0, and indicate if the test is left tailed (<), right taile (>), or two tailed ($\neq$), then click Calculate.

Standard Deviation (s)

Sample Size (n):

Sample Mean ($\bar{x}$):

Choose the test

- $<$
- $>$
- $\neq$

Hypothesized Population Mean (μ_0):

Calculate

Output

Test Statistic (t)

p-value

Scientific Calculator

14: Hypothesis Test for a Population Mean Given Data

Hypothesis Test for a Population Mean With Data

This calculator performs the hypothesis test for a population mean given statistics. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here (coming up).

Input

Type in the values from the data set separated by commas, for example, 2,4,5,8,11,2. Then choose the left, right, or two-tailed test and the hypothesized mean. Finally hit Calculate and the sample mean, then the test statistic and the p-value will be shown.

Data:

Choose the test

$<$

$>$

$\neq$

Hypothesized Population Mean (μ_0):

Calculate

Output

Sample Mean ($\bar{x}$):

Standard Deviation (s):

Test Statistics (t)

p-value:

Scientific Calculator

15: Two Independent Proportions Comparison

Two Independent Proportions Comparison

This calculator performs the hypothesis test and also constructs a confidence interval for $p_1 - p_2$ given statistics. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here.

Input

Enter the sample size and the number of successes for each sample, choose the test, and enter the confidence level then hit Calculate. The test statistic, p-value, and the boundaries of the confidence interval will be shown. Be sure to enter the confidence level as a decimal, e.g., enter 95% as 0.95.

	Sample Size (N)	Number of Successes (X)
First Sample		
	choose a test ◉ < ○ > ○ ≠	
Second Sample		

Confidence Level (enter a decimal):

.95

Calculate

Output

Test Statistics (z):	p-value:	Lower Bound:	Upper Bound:

Scientific Calculator

16: Two Independent Sample Means Comparison Given Statistics

Two Independent Samples with statistics Calculator

This calculator performs the hypothesis test and also constructs a confidence interval for $\mu_1 - \mu_2$ for two population means given statistics. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here (coming up).

Input

Enter the statistics, the tail type, and the confidence level and hit Calculate and the test statistic, p-value, and the boundaries for the confidence interval. Be sure to enter the confidence level as a decimal, e.g., 95% should be entered as 0.95.

	Sample Size	Sample Mean	Sample Standard Deviation
First Sample			
	choose a test < > ≠		
Second Sample			

Confidence Level (Enter a Decimal):

.95

Calculate

Output

Test Statistics (t):	p-value:	Lower Bound:	Upper Bound:

Scientific Calculator

Back to the Calculator Menu

17: Two Independent Sample Means Comparison Given Data

Two Independent Samples Means Comparison Given Data

This calculator performs the hypothesis test and also constructs a confidence interval for $\mu_1 - \mu_2$ given data. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here (coming up).

Input

Type in the values from the two data sets separated by commas, for example, 2,4,5,8,11,2. Enter the statistics, the tail type, and the confidence level then hit Calculate. The test statistic, p-value, and the boundaries for the confidence interval. Be sure to enter the confidence level as a decimal, e.g., 95% should be entered as 0.95.

Data 1:

choose a test

<

>

≠

Data 2:

Confidence Level (enter a decimal):

0.95

Calculate

Output

Test Statistics (t):	p-value	Lower bound	Upper Bound

Scientific Calculator

18: Two Dependent Sample Means Comparison Given Data

Two dependent Samples Means Comparison Given data

This calculator performs a hypothesis test and creates a confidence interval for two dependent sample means given the data sets. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here (coming up).

Input

Type in the values from the two data sets separated by commas, for example, 2,4,5,8,11,2. Then enter the tail type and the confidence level and hit Calculate and the test statistic, t, the p-value, p, the confidence interval's lower bound, LB, the upper bound, UB, and the data set of the differences will be shown. Be sure to enter the confidence level as a decimal, e.g., 95% has a CL of 0.95.

Data1:

choose a test

<

>

≠

Data2:

Confidence Level (enter a decimal):

0.95

Calculate

Output

Test Statistics (t):	p-value:	Lower Bound:	Upper Bound:

Scientific Calculator

19: Chi-Square Test for Goodness of Fit

Chi-Square Test for Goodness of Fit

This calculator performs the Chi-Square test for goodness of fit. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here.

Input

Type in the values from the observed and expected sets separated by commas, for example, 2,4,5,8,11,2. Then hit Calculate, the test statistic and the p-value will be shown.

Observed:

Expected:

Output

Calculate

Test Statistics (χ^2):	p-value:

Scientific Calculator

20: Chi-Square Test for Independence

Under Reconstruction

χ^2 test for independence calculator

Enter in the observed values and hit Calculate and the χ^2 test statistic and the p-value will be calculated for you. Leave blank the last rows and columns that don't have data values.

	A	B	C	D
First				
Second				
Third				
Fourth				
Fifth				
Sixth				
Seventh				
Eighth				
Ninth				
Tenth				

Calculate

χ^2:	p

Scientific Calculator

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21: One-Way ANOVA

One-Way ANOVA

This calculator performs the one-way ANOVA. Please report the error to Dr. Jessica Kuang at jkuangATvcccd.edu.

To learn how to use this calculator, please watch a short video here (coming up).

Input

Enter the data values separated by commas, for example: 3,4,7,9,-2,8. Then hit Calculate and the F-statistic and the p-value will be generated for you. Leave the extra rows blank if not needed.

Data1:

Data2:

Data3:

Data4:

Data5:

Data6:

Data7:

Data8:

Calculate	Test Statistics (F):	p-value:

Scientific Calculator

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Overview

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